

ANABAR, Russian Federation

WMO: 216080

Lat: 73.217N

Lon: 113.500E

Elev: 13

StdP: 101.17

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-46.2	-44.7	-47.2	0.0	-46.0	-45.8	0.0	-44.5	15.3	-20.7	14.2	-24.5	3.7	120	1.103

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	19.1	15.7	16.9	13.9	14.7	12.5	16.0	18.7	14.2	16.6	12.7	14.6	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.5	10.3	18.1	12.9	9.3	15.6	11.4	8.4	14.2	45.0	18.5	40.0	16.6	36.0	14.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.4	10.9	DB	-47.4	22.5	2.0	3.0	-48.8	24.6	-49.9	26.4	-51.0	28.1	-52.4	30.3	(4)
(5)			WB	-47.4	17.9	2.0	2.6	-48.8	19.8	-49.9	21.3	-51.0	22.8	-52.4	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.1	-32.8	-33.6	-26.3	-17.2	-7.7	4.6	8.7	8.2	2.2	-10.1	-24.2	-29.9
	DBStd	16.73	7.77	7.24	8.44	6.15	5.23	4.66	4.07	3.64	3.42	6.20	7.38	6.80
	HDD10.0	8488	1326	1221	1124	817	549	176	74	79	235	624	1027	1237
	HDD18.3	11459	1584	1454	1382	1067	808	413	301	313	484	882	1277	1496
	CDD10.0	72	0	0	0	0	0	13	34	24	1	0	0	0
	CDD18.3	2	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	14	0	0	0	0	0	1	13	0	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	5.3	4.9	4.7	5.0	5.6	5.7	5.7	5.6	5.6	5.5	4.9	5.3	5.2
Precipitation	PrecAvg	281	17	12	15	16	24	21	30	40	36	33	22	17
	PrecMax	395	54	28	26	41	47	43	64	80	86	58	44	39
	PrecMin	163	3	1	2	0	4	5	10	16	10	11	8	7
	PrecStd	51	11	6	6	9	9	9	13	18	19	10	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.5	-14.0	-3.3	0.4	4.1	21.5	25.8	20.2	13.6	1.2	-2.9	-8.8
		MCWB	-4.6	-14.4	-3.5	-0.8	2.7	16.9	20.3	16.2	12.0	1.0	-3.1	-8.9
	2%	DB	-15.3	-16.2	-5.7	-2.5	2.1	17.9	20.3	18.0	10.1	0.2	-6.2	-14.5
		MCWB	-15.5	-16.4	-6.2	-3.2	1.0	15.0	16.6	15.2	9.2	-0.1	-6.3	-14.7
	5%	DB	-18.0	-20.0	-8.3	-4.9	1.3	14.4	17.2	16.2	8.2	-1.3	-10.6	-17.2
		MCWB	-18.0	-20.2	-8.5	-5.3	0.4	11.8	14.1	13.3	7.4	-1.6	-10.8	-17.3
	10%	DB	-21.5	-22.6	-12.3	-7.3	0.1	11.4	15.0	14.0	6.5	-2.6	-13.7	-19.8
		MCWB	-21.6	-22.7	-12.6	-7.6	-0.5	9.6	12.7	12.2	5.8	-2.9	-14.0	-19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.7	-14.3	-3.5	-0.8	3.2	17.4	19.9	16.9	13.1	1.1	-3.0	-8.9
		MCDB	-4.6	-14.1	-3.3	0.2	3.9	20.1	25.3	19.6	13.8	1.1	-3.0	-8.7
	2%	WB	-15.4	-16.4	-6.1	-3.1	1.2	14.6	16.6	15.2	9.3	-0.1	-6.4	-14.6
		MCDB	-15.4	-16.2	-5.7	-2.7	1.9	17.7	19.5	17.8	9.7	0.1	-6.2	-14.5
	5%	WB	-18.0	-20.1	-8.6	-5.3	0.4	12.0	14.5	13.7	7.6	-1.5	-10.7	-17.3
		MCDB	-18.0	-20.0	-8.3	-4.9	1.1	14.3	17.0	15.6	8.4	-1.3	-10.5	-17.2
	10%	WB	-21.6	-22.7	-12.6	-7.6	-0.5	9.6	12.9	12.3	5.9	-2.8	-14.0	-19.9
		MCDB	-21.4	-22.6	-12.3	-7.3	0.2	11.4	15.0	14.0	6.4	-2.6	-13.7	-19.7
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.2	7.4	8.6	5.7	5.7	6.6	6.0	3.7	5.4	6.1	6.0
		MCDBR	8.9	10.1	10.2	10.6	6.2	11.2	9.7	9.5	7.2	5.2	8.7	8.7
		MCWBR	8.8	9.9	10.0	10.4	5.6	8.6	6.8	6.9	6.0	5.1	8.6	8.5
	5% WB	MCDBR	8.9	9.7	10.2	10.9	6.1	11.3	9.7	9.2	7.3	5.1	8.8	8.4
		MCWBR	8.8	9.5	10.0	10.8	5.7	8.8	7.1	7.2	6.2	5.1	8.8	8.4
Clear-Sky Solar Irradiance	taub	N/A	0.184	0.278	0.309	0.335	0.341	0.372	0.369	0.297	0.193	N/A	N/A	
	taud	N/A	1.901	2.020	2.063	2.105	2.359	2.341	2.310	2.394	1.946	N/A	N/A	
	Ebn at Noon	0	374	639	778	813	833	782	720	664	328	0	0	
	Edh at Noon	0	27	74	106	119	97	96	86	54	23	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.13	1.49	4.02	5.86	5.76	4.88	2.92	1.29	0.29	0.00	0.00	
	RadStd	0.00	0.01	0.07	0.17	0.44	0.48	0.35	0.40	0.15	0.04	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page